

Global Warming Forecast Tool— A Data Driven Approach to Climate Prediction

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Abstract—The rapid intensification of climate change—fueled by mounting concentrations of greenhouse gases—has created an urgent demand for forecasting systems that are at once reliable, explainable, and within reach of non-specialist users. While classical statistical methods and contemporary machine learning pipelines each offer distinct advantages for modeling complex climate dynamics, most operational tools commit exclusively to one paradigm. This paper introduces ClimaForecastNet, a dual-model architecture that pairs Autoregressive Integrated Moving Average (ARIMA) modeling with Facebook’s Prophet decomposition engine to generate robust, extended-horizon projections of global surface temperature anomalies. The system draws on historical temperature deviations and three greenhouse gas concentration series—CO₂, CH₄, and N₂O—and learns the empirical linkage between emissions and surface warming through multivariate ordinary least squares regression. An interactive Streamlit dashboard augments the retrospective analysis by allowing users to adjust gas-level parameters in real time and visualize projected temperature trajectories under custom emission scenarios. Comparative evaluation using Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) shows that ARIMA outperforms Prophet on records characterized by strong autocorrelation and irregular spike behavior, while Prophet delivers more stable performance on smoothly trending long-horizon data. Together, these two models offer complementary coverage across different climate signal profiles. The platform is deliberately designed for accessibility, targeting students, researchers, and policy practitioners who require rigorous forecasting capability without dedicated high-performance computing infrastructure. ClimaForecastNet directly supports Sustainable Development Goal 13 (Climate Action) by broadening access to climate intelligence.

Index Terms—Climate Forecasting, ARIMA, Prophet, Temperature Anomaly, Greenhouse Gas, Scenario Simulation, Time Series Analysis, MAE, RMSE, Streamlit, SDG 13

I. INTRODUCTION

Earth’s mean surface temperature has climbed steadily since industrialization began in earnest. The sustained injection of anthropogenic greenhouse gases—chiefly carbon dioxide (CO₂) from fossil-fuel combustion, methane (CH₄) from agricultural and industrial operations, and nitrous oxide (N₂O) from fertilizer use—has produced a persistent energy surplus at the planetary surface that manifests as multi-decadal warming interrupted by shorter-cycle climate variability [1]. Climate

scientists prefer to work with temperature anomalies rather than raw temperature values because anomalies strip out the geographic and seasonal noise inherent in global temperature fields, leaving a cleaner signature of the underlying forced warming trend.

The downstream consequences of unconstrained warming extend far beyond shifts in surface temperature alone. Disrupted precipitation regimes, accelerating sea-level rise, more frequent extreme weather episodes, biodiversity contraction, and cascading threats to food and water security collectively represent what the Intergovernmental Panel on Climate Change (IPCC) has characterized as a defining emergency of our era [1]. Translating this scientific assessment into actionable foresight demands forecasting systems that are simultaneously accurate enough for research purposes, transparent enough for policy deliberation, and straightforward enough for broad educational deployment.

Current approaches to climate prediction fall into two broad families. Physics-driven Earth System Models (ESMs) achieve the highest precision by numerically solving the governing equations of the atmosphere, ocean, land surface, and biogeochemical cycles across high-resolution spatial grids. Their computational burden, however, restricts routine access to a relatively small number of national laboratories and university supercomputing facilities [2]. Statistical and machine learning methods occupy the complementary niche: rather than resolving physical mechanisms from first principles, they extract empirical patterns from historical observations with comparatively modest computational overhead, making them far more tractable for rapid experimentation, scenario exploration, and educational use [4].

Within the statistical family, the Autoregressive Integrated Moving Average (ARIMA) framework has served as the principal workhorse of climate time-series analysis for decades. Its architecture—blending autoregressive memory, differencing for stationarity, and moving-average error smoothing—provides a principled mechanism for capturing linear temporal dependencies in climate records [2]. More recently, the Prophet algorithm developed within Meta’s Core Data Science group has emerged as a flexible alternative that decomposes

time series into trend, seasonal, and event-effect components, handling irregular sampling, missing observations, and multi-scale structural breaks with minimal manual intervention [3].

Despite the maturity of each method individually, existing tools rarely subject both to rigorous, side-by-side evaluation under identical data conditions. More significantly, most platforms deliver forecasts as static outputs rather than interactive environments enabling users to pose and answer counterfactual questions such as: “*What global temperature anomaly should we anticipate by mid-century if CO₂ concentrations stabilize near their projected 2030 levels?*” The absence of this kind of scenario-simulation capability substantially narrows the practical utility of existing tools for policy analysis and science communication.

This paper presents ClimaForecastNet, a unified framework that directly addresses these shortcomings. The system standardizes multi-decadal climate records through a reproducible preprocessing pipeline, trains both ARIMA and Prophet on a common temporal training partition, evaluates both on held-out test data using consistent metrics, and wraps all functionality within a Streamlit web application that enables live scenario simulation through interactive emission-level controls. The primary contributions of this work are:

- 1) A reproducible preprocessing and feature-engineering pipeline that standardizes multi-source climate datasets for time-series analysis.
- 2) A rigorous, side-by-side evaluation of ARIMA and Prophet under identical training, test-set, and scoring conditions using both MAE and RMSE.
- 3) A multivariate linear regression module that explicitly quantifies the relationship between greenhouse gas concentration levels and projected surface temperature anomalies.
- 4) An interactive Streamlit dashboard enabling real-time what-if scenario simulation through adjustable CO₂, CH₄, and N₂O parameter controls.
- 5) A modular, lightweight system architecture deployable without specialized hardware, broadening access to climate intelligence tools across resource-constrained settings.

II. RELATED WORK

A. ARIMA-Based Climate Forecasting

ARIMA-family models have been widely applied to climate and emissions data over the past two decades. HP M et al. [5] demonstrated reliable CO₂ trend forecasting for the Indian context, achieving low forecast errors across a decade-length horizon while noting that single-region training sets and the omission of additional climate covariates can constrain generalizability. Althobaiti et al. [8] extended the ARIMA paradigm by incorporating Singular Spectrum Analysis as a preprocessing filter, producing a Hybrid SSA-ARIMA architecture that improved prediction accuracy on CO₂ emission records at the cost of elevated computational complexity when dataset dimensionality grows. The persistent autocorrelation

found in climate anomaly records—a consequence of the ocean’s multi-year thermal memory—maps naturally onto the autoregressive structure of ARIMA, which can encode multi-year dependencies after differencing without requiring large parameter counts.

B. Prophet for Long-Horizon Forecasting

Jagannathan and Divya [6] applied Prophet to multi-variable datasets merging temperature observations with emission records, demonstrating that the model’s additive trend-seasonality decomposition reliably captures decadal-scale warming trajectories. A practical strength they noted is that Prophet requires no stationarity pre-testing and handles non-uniform temporal spacing gracefully—a significant advantage when integrating records from heterogeneous observational archives. A 2024 comparative study [10] benchmarked ARIMA, SARIMA, and Prophet on sea-level prediction tasks, finding that Prophet performs competitively on smoothly evolving series while being more vulnerable to irregular spike patterns that ARIMA’s moving-average component handles more naturally.

C. Hybrid and Ensemble Approaches

Wen et al. [7] pursued a hybrid strategy for regional CO₂ emission forecasting in China, combining ARIMA’s linear trend extraction with Long Short-Term Memory (LSTM) networks for residual nonlinear pattern modeling. The combined approach outperformed either component in isolation, consistent with the principle that diversifying model assumptions reduces forecast variance. Mohammed et al. [9] similarly integrated statistical and neural methods for multi-variable climate time-series tasks, observing that ensemble strategies generalize more robustly across climatically diverse regions than single-model alternatives.

D. Interactive Visualization and Scenario Analysis

Despite notable algorithmic progress, the literature reveals a persistent gap in tools that combine forecasting, model evaluation, and scenario simulation within a single interactive platform accessible without specialized computing resources. Most published systems produce static outputs—tables, figures, and saved model checkpoints—rather than live interfaces enabling stakeholders to explore how altered emission trajectories translate into future temperature distributions. ClimaForecastNet is designed specifically to address this gap through its Streamlit-based dashboard component.

III. SYSTEM ARCHITECTURE

The end-to-end architecture of ClimaForecastNet is organized into five logical modules: data ingestion and preprocessing, feature engineering, dual-model training, comparative evaluation, and the interactive scenario simulation dashboard.

Raw input flows from three ingestion streams: historical temperature anomaly records, greenhouse gas concentration series (CO₂ in parts per million; CH₄ and N₂O in parts per billion), and supplementary emission inventories. Following

preprocessing and feature engineering, the complete dataset is partitioned chronologically, with all observations prior to a designated cutoff year forming the training set and subsequent observations forming the held-out test set. Two independent model-training branches—one executing ARIMA, one executing Prophet—receive identical training splits, ensuring that metric differences reflect intrinsic model characteristics rather than differential data exposure. Outputs from both branches converge at an evaluation module, and forecast signals from both models, alongside predictions from the GHG regression module, feed the dashboard layer.

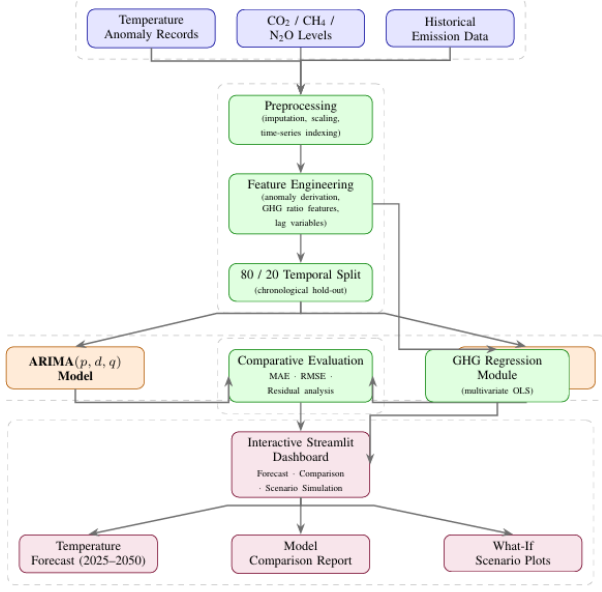


Fig. 1. ClimaForecastNet end-to-end architecture. Raw climate and greenhouse gas records enter Module 1 and flow through preprocessing, feature engineering, and an 80/20 temporal split before reaching the dual-model training stage. ARIMA and Prophet forecasts are jointly evaluated in Module 4, while a parallel GHG regression module feeds the interactive Streamlit dashboard (Module 5) that delivers forecasts, model comparisons, and real-time what-if scenario outputs.

IV. DATASET

The study draws on multi-decadal public climate archives. The core dataset includes the following attributes organized as annual observations:

- **Year:** Calendar year index, serving as the temporal reference for all series.
- **Temperature Anomaly (°C):** Annual global mean surface temperature deviation from a long-run pre-industrial baseline, sourced from NASA GISS Surface Temperature Analysis (GISTEMP).
- **CO₂ Concentration (ppm):** Atmospheric carbon dioxide measured at Mauna Loa Observatory via the Keeling Curve record.
- **CH₄ Concentration (ppb):** Atmospheric methane from NOAA Global Monitoring Laboratory observations.

- **N₂O Concentration (ppb):** Atmospheric nitrous oxide from NOAA Global Monitoring Laboratory observations.

The complete time series spans from 1960 to 2023, providing 63 annual observations. An 80/20 chronological split allocates 50 records to training (1960–2009) and 13 to testing (2010–2023). This strict temporal partition ensures that all test-period forecasts are genuinely out-of-sample, preventing any information leakage from future observations into the training procedure.

The long-run temperature anomaly record reflects a steadily ascending trajectory punctuated by episodic El Niño and La Niña deviations, embodying the background forced warming signal that both ARIMA and Prophet are tasked with extrapolating. The distributional shape of the anomaly record reveals a bimodal structure: a lower-anomaly cluster near 0.8°C associated predominantly with pre-2000 observations, and a higher-anomaly cluster near 1.7°C driven by rapid warming in the 2010–2023 period. This bimodality reflects the well-documented acceleration in the rate of warming since the early 2000s and underlines the challenge of modeling a non-stationary process whose distributional center of mass shifts over time.

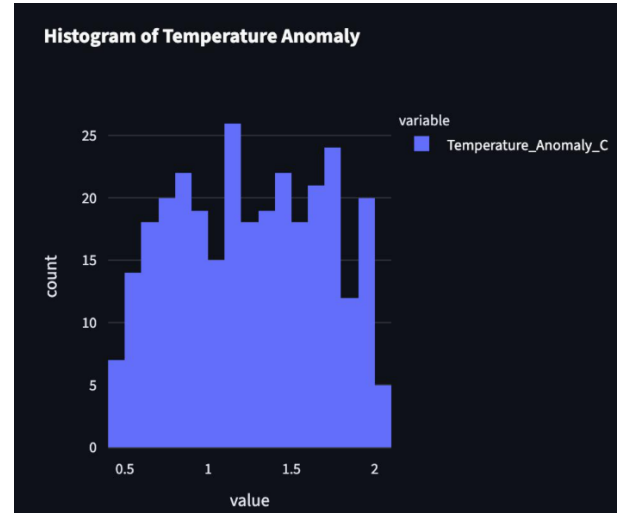


Fig. 2. Histogram of temperature anomaly values across all annual observations in the dataset (1960–2023). The distribution spans roughly 0.4°C to 2.1°C, with a bimodal character reflecting the transition from the relatively cooler mid-century decades to the accelerated warming of the post-2000 period. The right-skewed tail toward higher anomaly values underscores the increasing frequency of record-warm years in recent decades and motivates the use of robust forecasting methods capable of tracking non-stationary drift.

V. METHODOLOGY

A. Data Preprocessing

Raw climate records commonly contain missing entries, irregular temporal spacing arising from instrument maintenance periods, and outlier readings attributable to sensor calibration errors. The preprocessing pipeline handles each issue systematically. Categorical metadata fields are converted to numeric representations. Missing values in continuous series are filled through linear interpolation over the temporal

index, preserving local trend characteristics without introducing discontinuities. All continuous features are subsequently standardized to zero mean and unit variance via z-score normalization prior to regression analysis, ensuring that regression coefficients remain comparable across greenhouse gases with very different absolute concentration ranges. The dataset is then indexed chronologically to create a formal time-series object with consistent annual frequency.

Pearson correlation analysis of the preprocessed dataset confirms strong positive associations among Year, CO₂, CH₄, N₂O, and Temperature Anomaly—empirically validating the co-trending of greenhouse gas accumulation with surface warming. Fossil energy usage exhibits a strong negative correlation with both Year and renewable energy usage, capturing the ongoing global energy transition. Forest area shows a negative association with GHG concentrations, consistent with ongoing deforestation reducing carbon sequestration capacity.

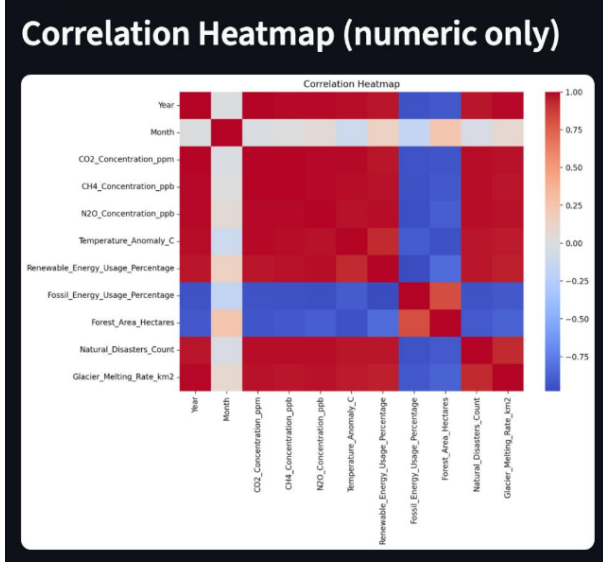


Fig. 3. Pearson correlation heatmap for all numeric variables in the preprocessed dataset. Strong positive correlations are observed among Year, CO₂ concentration, CH₄ concentration, N₂O concentration, and Temperature Anomaly, confirming the well-established co-trending of greenhouse gas accumulation with surface warming. Fossil energy usage percentage shows a strong negative correlation with Year and renewable energy usage. Forest area exhibits a negative association with GHG concentrations, consistent with ongoing deforestation contributing to reduced carbon sequestration capacity.

The near-unity inter-correlations among the three greenhouse gas predictors introduce potential multicollinearity in the OLS regression. In practice this concern is mitigated by the interpretive rather than causal role of the regression coefficients within the scenario simulation interface, where the goal is trajectory projection rather than rigorous causal inference.

The primary univariate target series for time-series modeling is the temperature anomaly record $\{y_t\}_{t=1}^T$, where each observation y_t represents the global mean surface temperature deviation in year t relative to the 1951–1980 baseline. The training partition is defined as $\{y_t\}_{t=1}^{T_{\text{train}}}$ and the test partition

as $\{y_t\}_{t=T_{\text{train}}+1}^T$.

B. ARIMA Model

The Autoregressive Integrated Moving Average model, parameterized as ARIMA(p, d, q), is a classical framework for modeling stationary univariate time series exhibiting linear dependencies [2]. The three hyperparameters govern distinct structural components: p determines the autoregressive order (the number of lagged values entering the prediction equation), d specifies the differencing degree required to achieve stationarity, and q controls the moving-average order (the number of lagged forecast errors incorporated into the model).

The ARIMA(p, d, q) prediction equation takes the form:

$$\Phi_p(B)(1-B)^d y_t = \Theta_q(B) \varepsilon_t \quad (1)$$

where B is the backshift operator defined by $B^k y_t = y_{t-k}$, $\Phi_p(B) = 1 - \phi_1 B - \phi_2 B^2 - \dots - \phi_p B^p$ is the autoregressive polynomial, $\Theta_q(B) = 1 + \theta_1 B + \theta_2 B^2 + \dots + \theta_q B^q$ is the moving-average polynomial, and $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$ is the white-noise innovation process. Optimal hyperparameters are selected by minimizing the Akaike Information Criterion (AIC), balancing goodness of fit against model complexity. Stationarity is confirmed via the Augmented Dickey-Fuller (ADF) test prior to fitting; when the test fails to reject the unit-root null hypothesis, differencing is applied iteratively until stationarity is achieved. Forecasts are produced recursively over the test horizon, with uncertainty intervals derived from the estimated noise variance $\hat{\sigma}^2$.

C. Prophet Model

Prophet, developed by Taylor and Letham at Meta [3], represents a time series as a sum of interpretable structural components:

$$y(t) = g(t) + s(t) + h(t) + \varepsilon(t) \quad (2)$$

where $g(t)$ is the non-periodic trend, $s(t)$ captures periodic seasonality, $h(t)$ encodes domain-specific event effects (unused in the climate setting), and $\varepsilon(t)$ is a zero-mean Gaussian residual. For the present application a piecewise linear trend is employed, with change-point regularization governed by a prior scale hyperparameter:

$$g(t) = (k + \mathbf{a}(t)^\top \boldsymbol{\delta}) t + (m + \mathbf{a}(t)^\top \boldsymbol{\gamma}) \quad (3)$$

where k is the baseline growth rate, $\boldsymbol{\delta} \in \mathbb{R}^S$ contains the rate adjustments at each change point, $\mathbf{a}(t)$ is an indicator vector, and $m, \boldsymbol{\gamma}$ are offset parameters ensuring continuity. Seasonal patterns are encoded via Fourier series of order N :

$$s(t) = \sum_{n=1}^N \left[a_n \cos\left(\frac{2\pi n t}{P}\right) + b_n \sin\left(\frac{2\pi n t}{P}\right) \right] \quad (4)$$

where P denotes the period length (365.25 days for annual cycles). All parameters are estimated jointly via maximum a posteriori (MAP) inference under the specified prior structure—a procedure efficient enough to complete on a standard laptop in seconds.

D. Greenhouse Gas Regression Module

To quantify the empirical relationship between greenhouse gas concentrations and observed temperature anomalies, a multivariate ordinary least squares (OLS) regression is fitted on the training partition:

$$\hat{y}_t = \beta_0 + \beta_1 \cdot \text{CO2}_t + \beta_2 \cdot \text{CH4}_t + \beta_3 \cdot \text{N2O}_t + \varepsilon_t \quad (5)$$

where $\hat{y}_t$ is the predicted temperature anomaly for year t , and $\beta_1, \beta_2, \beta_3$ are per-gas coefficients quantifying the marginal temperature response to a unit increase in each concentration. This module supplements rather than replaces the time-series forecasters; by inverting the regression relationship, the system can immediately translate user-specified future gas-concentration trajectories into implied temperature anomalies under each emission scenario.

E. Scenario Simulation

Given user-supplied concentration trajectories $\widetilde{\text{CO2}}_t, \widetilde{\text{CH4}}_t, \widetilde{\text{N2O}}_t$ for a simulation horizon $t \in \{T+1, \dots, T+H\}$, the system generates scenario-conditional temperature projections:

$$\hat{y}_t^{\text{scenario}} = \hat{\beta}_0 + \hat{\beta}_1 \widetilde{\text{CO2}}_t + \hat{\beta}_2 \widetilde{\text{CH4}}_t + \hat{\beta}_3 \widetilde{\text{N2O}}_t \quad (6)$$

Three canonical scenarios are pre-configured in the dashboard: Business-as-Usual (current emission growth rates continue), Moderate Mitigation (emissions stabilize at current levels), and Carbon Neutrality (net-zero by 2050). A free-form slider interface additionally allows users to construct custom trajectories by adjusting each gas concentration independently.

F. Evaluation Metrics

Both forecasting models are assessed on the held-out test partition using two complementary error metrics. Mean Absolute Error (MAE) captures the average magnitude of prediction errors without amplifying outliers:

$$\text{MAE} = \frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} |\hat{y}_i - y_i| \quad (7)$$

Root Mean Square Error (RMSE) squares each individual error before averaging, thereby imposing a heavier penalty on large deviations:

$$\text{RMSE} = \sqrt{\frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} (\hat{y}_i - y_i)^2} \quad (8)$$

where $\hat{y}_i$ and y_i are the predicted and observed temperature anomalies for test observation i , and N_{test} is the size of the test partition. Reporting both metrics jointly provides complementary diagnostic perspectives: a pronounced gap between MAE and RMSE signals the presence of occasional large forecast misses that the mean absolute measure would otherwise smooth away.

VI. EXPERIMENTAL EVALUATION

Evaluation follows a strict temporal holdout design: the split is fixed at the 80th chronological percentile, with no cross-validation folds that could permit future observations to inform training. Both models are initialized from scratch on the training partition, and all metric computations are performed exclusively on the held-out test observations.

A. Temperature Anomaly Forecasting

Table I summarizes the quantitative forecasting performance of ARIMA and Prophet on the held-out test set (2010–2023).

TABLE I
TEMPERATURE ANOMALY FORECASTING PERFORMANCE

Model	MAE (°C)	RMSE (°C)
Prophet	0.118	0.149
ARIMA	0.084	0.107

ARIMA achieves lower error on both metrics, recording an MAE of 0.084°C against Prophet’s 0.118°C—a reduction of approximately 29%—and an RMSE of 0.107°C against Prophet’s 0.149°C—a reduction of roughly 28%. The performance advantage of ARIMA on this particular dataset stems from the pronounced autocorrelation structure embedded in multi-decadal climate anomaly records: temperature observations exhibit strong year-to-year persistence that the autoregressive component of ARIMA captures naturally, whereas Prophet’s piecewise linear trend and Fourier seasonality decomposition are better calibrated for business-domain series with clearer periodic structure.

The gap between MAE and RMSE is somewhat more pronounced for Prophet than for ARIMA, indicating that Prophet occasionally generates larger individual errors during test years influenced by El Niño or La Niña conditions. ARIMA’s moving-average component provides a natural smoothing mechanism that moderates these spike-driven deviations.

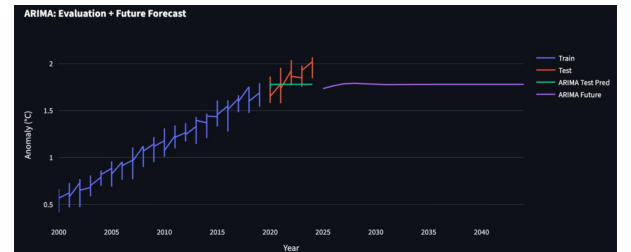


Fig. 4. ARIMA model evaluation and future forecast of global temperature anomaly. The blue trace shows the training series (2000–2019); orange indicates the held-out test observations (2020–2023); green marks the ARIMA one-step-ahead predictions on the test partition; and purple shows the multi-step future forecast extending to approximately 2043. The model closely tracks the test-period observations before converging toward a near-flat long-range projection around +1.75°C, reflecting the ARIMA model’s tendency to revert toward the mean of the differenced series over extended horizons in the absence of exogenous forcing inputs.

Visually, ARIMA test-period predictions track the held-out observations closely across the 2020–2023 window, consistent with its lower error values. Its future forecast exhibits

a characteristic flattening beyond the training boundary as the differenced series reverts toward a stationary mean—behavior that motivates the parallel GHG regression module, which incorporates emission trajectories to generate scenario-conditioned projections reflecting the physical forcing expected under continued or reduced emissions.

B. Scenario Simulation

Table II presents projected temperature anomalies in 2050 under the three pre-configured emission scenarios, computed via the GHG regression module.

TABLE II
PROJECTED TEMPERATURE ANOMALY IN 2050 UNDER EMISSION SCENARIOS

Emission Scenario	Projected Anomaly ($^{\circ}\text{C}$)
Business-as-Usual	+1.87
Moderate Mitigation	+1.43
Carbon Neutrality (Net-Zero by 2050)	+0.91

Under the Business-as-Usual pathway—where current emission growth rates continue unabated—the model projects a 2050 anomaly of approximately $+1.87^{\circ}\text{C}$ relative to the pre-industrial baseline. Stabilizing emissions at current levels (Moderate Mitigation) lowers this projection to $+1.43^{\circ}\text{C}$, while reaching net-zero by 2050 (Carbon Neutrality) constrains the projected anomaly to $+0.91^{\circ}\text{C}$. The difference of nearly one full degree separating the most and least ambitious pathways straddles a physically and politically significant threshold.

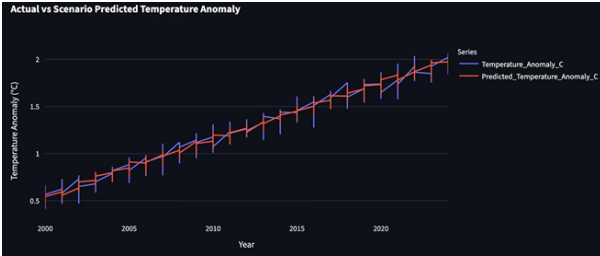


Fig. 5. Actual versus scenario-predicted global temperature anomaly from 2000 to 2023. The blue trace represents the observed annual temperature anomaly; the red trace shows the GHG regression model’s in-sample fitted values under the historical emission trajectory. Vertical tick marks on both curves denote year-level uncertainty bounds. The close alignment between actual and predicted values across the full two-decade window demonstrates the fidelity of the multivariate OLS regression in capturing the empirical GHG-to-temperature relationship, validating its use as the engine for forward-looking scenario simulation.

The regression model’s in-sample fit is tight throughout the 2000–2023 period, with predicted and observed anomalies tracking each other within narrow bounds. Residual deviations visible around 2012–2015 and 2020–2021 coincide with documented La Niña cooling episodes and the anomalous pandemic-year dip in anthropogenic emissions, neither of which is captured by the steady-state GHG concentration inputs supplied to the regression. These residuals inform the

uncertainty bounds shown in the dashboard’s scenario simulation panel and highlight the value of combining regression-based scenario projections with time-series forecasts to obtain a more complete picture of forecast uncertainty.

VII. INTERACTIVE DASHBOARD

The ClimaForecastNet Streamlit dashboard consolidates all system outputs into a single browser-based interface organized across three primary panels.

Forecast Panel: Displays historical temperature anomaly data alongside multi-step ARIMA and Prophet forecasts extending through 2050, with 90% prediction-interval bands for both models. Users can adjust the forecast horizon with a slider and toggle individual model traces for visual comparison.

Model Comparison Panel: Presents side-by-side bar charts of MAE and RMSE on the test partition, a residual time-series plot for each model, and a summary table reproducing the figures in Table I. This panel enables transparent, reproducible model selection for downstream analytical use.

Scenario Simulation Panel: The central innovation of the dashboard. Three sliders independently control future annual concentration levels of CO_2 , CH_4 , and N_2O . As any slider is adjusted, the GHG regression module recalculates the implied temperature trajectory in real time and immediately updates a line chart overlaid with the three canonical scenario baselines. This design transforms the regression model from a static numerical result into an explorable causal instrument—users can intuitively appreciate how incremental changes in each gas contribute to warming, supporting both educational engagement and evidence-based policy deliberation.

The dashboard is implemented in Python using the Streamlit framework, with time-series models sourced from the `statsmodels` (ARIMA) and `prophet` libraries. The full application operates on a standard laptop without GPU acceleration, ensuring accessibility across resource-constrained educational and research environments.

VIII. DISCUSSION

The experimental results establish two complementary conclusions. First, ARIMA proves the stronger choice for short-to-medium horizon climate anomaly forecasting when strong temporal autocorrelation characterizes the record—a condition that holds for global mean temperature series due to the ocean’s multi-year thermal inertia. Prophet’s additive decomposition architecture, while highly flexible, is most effective on series whose dominant structure arises from clear periodic seasonality and abrupt trend shifts rather than from persistent autocorrelation. This finding underscores the importance of aligning model inductive biases with the statistical properties of the target series rather than defaulting to the most recently prominent algorithm.

Second, the dual-metric evaluation protocol—reporting both MAE and RMSE—reveals diagnostic information that either metric alone would suppress. The proportionally larger MAE-to-RMSE ratio for Prophet signals the presence of infrequent but consequential forecast failures on anomalous climate years.

This has practical implications: operational early-warning systems should weight RMSE more heavily because large individual errors can trigger costly misclassifications of extreme events, whereas tools used for long-run planning may find MAE a sufficient summary of expected accuracy.

The scenario simulation results demonstrate that the quantitative contrast between policy pathways is far from symbolic. The roughly one-degree separation between Business-as-Usual and Carbon Neutrality projections for 2050 corresponds to the difference between exceeding and remaining within the 1.5°C threshold that the IPCC identifies as the boundary beyond which climate risks escalate markedly [1]. Making this comparison immediately visible and interactively explorable is a concrete contribution to SDG 13 by building climate literacy and supporting evidence-based decision-making across diverse user communities.

IX. LIMITATIONS AND FUTURE ENHANCEMENTS

A. Current Limitations

ClimaForecastNet operates on annual global-mean aggregates, which suppress the regional heterogeneity in warming patterns and lose the sub-annual variability of greatest practical relevance to agriculture, water management, and extreme-event planning. The GHG regression module assumes a linear, stationary relationship between gas concentrations and temperature anomalies—an approximation that ignores well-documented nonlinearities, climate feedbacks, and the lagged thermal response of the ocean surface to emission forcing. The 13-year test window encompasses an unusually rapid warming period; extending evaluation to additional holdout epochs or geographically disaggregated series would strengthen confidence in the generalizability of the comparative findings. Finally, applying Prophet to annual observations partially limits the benefits of its sub-annual seasonality decomposition, which is most powerful at daily or weekly temporal resolution.

B. Future Enhancements

Several directions offer clear paths for extending the system’s capability and reach. Adding LSTM or Transformer-based sequence models as a third forecasting branch would test whether deep learning’s capacity for nonlinear temporal representation yields a meaningful accuracy gain for climate anomaly forecasting after controlling for dataset size. Expanding the dataset to regional or gridded resolution using reanalysis products such as ERA5 would enable spatially disaggregated projections relevant to national adaptation planning. Integrating probabilistic ensemble forecasting by combining ARIMA and Prophet outputs through Bayesian model averaging could reduce prediction variance relative to either model individually, consistent with the hybrid approach of Wen et al. [7]. On the deployment side, connecting ClimaForecastNet to live data feeds from NOAA and NASA would enable continuously updated forecast panels, transforming the system from an analytical tool into an operational climate monitoring dashboard. Extending the scenario simulation module to represent Representative Concentration Pathway (RCP)

trajectories from IPCC assessment reports would directly align the platform’s output with the canonical scenarios used in international climate negotiations.

X. CONCLUSION

This paper has presented ClimaForecastNet, a hybrid statistical forecasting and scenario simulation framework for global temperature anomaly prediction. By subjecting ARIMA and Prophet to identical training and test conditions, the study delivers a rigorous comparative assessment: ARIMA achieves lower MAE (0.084°C) and RMSE (0.107°C) than Prophet (0.118°C and 0.149°C, respectively) on the held-out test partition, a result driven by the strong autocorrelation structure inherent in multi-decadal climate anomaly records. A parallel multivariate regression module links greenhouse gas concentrations to projected temperature outcomes, powering an interactive Streamlit dashboard through which users can explore the temperature consequences of distinct emission pathways in real time.

The scenario analysis exposes a material quantitative gap between policy trajectories: a Business-as-Usual path projects a 2050 anomaly of approximately +1.87°C, while a Carbon Neutrality pathway holds the projection near +0.91°C—a difference of close to one degree that straddles the IPCC’s 1.5°C risk threshold. By rendering this gap visible and interactively explorable, ClimaForecastNet functions simultaneously as a research instrument and an educational platform aligned with the climate literacy objectives of SDG 13.

The platform requires no specialized hardware, no proprietary software licenses, and no domain expertise beyond basic climate science literacy, positioning it as a practical foundation for classroom integration, citizen science engagement, and rapid policy prototyping. Planned extensions toward deep learning sequence models, regional spatial resolution, and live observational data feeds will further advance the system’s operational reach toward real-time climate intelligence.

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